

EYEPIECES

Most telescopes are supplied with one or two eyepieces: one that gives a basic low magnification, or power; and the other providing a higher power. To increase magnification further you need additional eyepieces, but there is also a limit to the power that any telescope can tolerate, often given as twice its aperture in millimeters. For example, the limit of a 130 mm telescope is 260. As the power is increased, the field of view usually decreases, the image dims, any atmospheric turbulence (called the “seeing”) is emphasized, and it becomes harder to keep objects within the field of view. One way to increase the power of a set of eyepieces is to place a Barlow lens between the telescope and the eyepiece. This lens typically doubles the power of each eyepiece, giving you a wider range of magnifications from a small set of eyepieces.



EYEPIECES

Telescope eyepieces are available in a range of focal lengths, with the highest figure giving the lowest magnification. The optical design varies, some combining a very wide apparent field of view with a high power.



NAKED-EYE VIEW

TELESCOPE VIEW

STAR DIAGONAL

This is a device often used with refractor and catadioptric telescopes to improve observing position, but it also reverses the image.



FILTERING OUT LIGHT POLLUTION

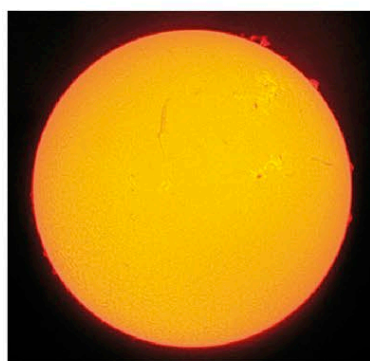
Sodium street lamps emit yellow light with a narrow range of wavelengths, making the sky glow orange (above). A light-pollution filter can cut it out while leaving the light from distant stars unaffected (right).



ANTI-LIGHT-POLLUTION FILTER



SOLAR TELESCOPE



THE SUN'S SURFACE

This solar telescope view of the Sun shows granulation and sunspots; bright areas called faculae; prominences at the Sun's edge, or lim; and strandlike filaments seen against the Sun's bright surface.

SOLAR TELESCOPES

The Sun is a fascinating object to observe with constantly changing features, but it is also the most dangerous, because it is so bright that even a momentary view through a telescope can blind the viewer. Specialized filters are available that reduce the brightness of the incoming light, which must be done because light enters the tube rather than at the eyepiece, where the light is focused. Only filters specifically designed for the purpose should be used because other dense material may transmit harmful infrared light. Many of the Sun's most fascinating features are visible only in the deep red hydrogen-alpha wavelength emitted by hydrogen gas. Filters that only transmit this light are very expensive, so even a basic solar telescopes can cost as much as a digital SLR camera. Specialized instruments called solar telescopes are also available. These reveal fascinating detail on the surface of the Sun, as well as the prominences around its edge.

STAR PARTY

Amateur astronomers gather in dark-sky areas at what are often called star parties. Only red lights are allowed, because they interfere less with night vision than lights of any other color.